

7/10/2026 - FINISH SERVICE AREA + RECURSIVE FEASIBILITY

$$P_{ik}^{ESS} = P_{ik}^{ch} - P_{ik}^{dis}, \quad i = 0, 1, \dots, N-1$$



$$P_{ik}^{ch}, P_{ik}^{dis} \geq 0, \quad i = 0, \dots, N-1$$

LOGICAL CONSTR.: CHARGING AND DISCHARGING MUTUALLY EXCLUSIVE
EVENT 1 EVENT 2

$$\bullet \left(\delta_{ik}^{ch} = 1 \Leftrightarrow p_{ik}^{ch} > 0 \right), \text{ INDICATOR CONSTRAINT}$$

$i = 0, \dots, W-1$

$$\bullet \delta_{ik}^{ch} \in \{0, 1\}, i = 0, \dots, W-1$$

$$\bullet \left(\delta_{ik}^{dis} = 1 \Leftrightarrow p_{ik}^{dis} > 0 \right), \forall i$$
$$\bullet \delta_{ik}^{dis} \in \{0, 1\}, \forall i$$

MUTUALLY EXCLUSIVE EVENTS

$$\bullet \quad \underline{\delta_{ik}^{ch} + \delta_{ik}^{dis} \leq 1} \quad ; \quad i = 0, 1, \dots, N-1$$

$$\underline{\delta_{ik}^{ch} = 0} \longrightarrow P_{ik}^{ch} = 0$$

$$\delta_{ik}^{dis} = 0 \longrightarrow P_{ik} = 0$$

$$\left(\delta_{ik}^{ch} = 1 \right) \begin{array}{c} \xrightarrow{\quad} \\ \Leftrightarrow \end{array} \left(p_{ik}^{ch} > 0 \right)$$

|||

$$\left(\delta_{ik}^{ch} = 0 \right) \begin{array}{c} \xrightarrow{\quad} \\ \Leftrightarrow \end{array} \left(p_{ik}^{ch} = 0 \right)$$

$$\left(p_{ijk}^{ch} > 0 \right) \longrightarrow \left(\delta_{ijk}^{ch} = 1 \right)$$

$$\left(\delta_{ijk}^{ch} = 0 \right) \longrightarrow \left(p_{ijk}^{ch} = 0 \right)$$

RELAXED
INDICATOR
CONSTRAINT

$$\delta_{ijk}^{ch}, p_{ijk}^{ch}, M \gg 1, \leq 1$$

CHARGING INDICATOR CONSTRAINT

$$P_{ik}^{ch} \leq M \cdot \delta_{ik}^{ch}, \quad i = 0, 1, \dots, N-1$$

- $P_{ik}^{ch} > 0 \rightarrow \delta_{ik}^{ch} = 1$
- $P_{ik}^{ch} = 0 \rightarrow \delta_{ik}^{ch} \in \{0, 1\}$
- $\delta_{ik}^{ch} = 0 \rightarrow P_{ik}^{ch} = 0$

WE CAN TAKE $M = P^{ch, \max}$

$$\delta_{ik}^{ch} = 1 \rightarrow P \in [0, P^{ch, \max}]$$

ESS DYNAMICS WITH CH. AND DISCH. LOSSES

$$x_{i+1|k} = x_{i|k} + \underbrace{\eta^{ch} \cdot T \cdot P_{i|k}^{ch}}_{\text{CHARGED ENERGY}} - \underbrace{\frac{1}{\eta^{dis}} \cdot T \cdot P_{i|k}^{dis}}_{\text{DISCHARGED ENERGY}}$$

$$i = 0, 1, \dots, N-1$$

COMPLETE FORMULATION

$$\min \left\{ \sum_{i=0}^{N-1} \left[a (P_{i|k}^{POC})^2 + b (x_{i|k} - x^{ref})^2 \right] + b^T (x_{N|k} - x^{ref})^2 \right\}$$

SUBJECT TO :

KNOWN DISTURBANCES

- $P_{i|k}^{POC} = P_{i|k}^{ESS} + P_{i|k}^{PEV} - P_{i|k}^{RES}$, $i = 0, 1, \dots, N-1$ (POWER BALANCE)

- $P_{i|k}^{ch} \geq 0, P_{i|k}^{dis} \geq 0$, $i = 0, 1, \dots, N-1$ (DEFINITION CH/DISCH. POWERS)

- $P_{i|k}^{ESS} = P_{i|k}^{ch} - P_{i|k}^{dis}$, $i = 0, 1, \dots, N-1$ (ESS POWER BALANCE)

- $\delta_{i|k}^{ch} \in \{0, 1\}, \delta_{i|k}^{dis} \in \{0, 1\}$, $i = 0, 1, \dots, N-1$ (INDICATOR VARIABLES)

- $p_{ik}^{ch} \leq p^{ch, max} \cdot \underline{\delta_{ik}^{ch}}$, $i=0, 1, \dots, N-1$ (CH. INDICATOR CONSTRAINT)
- $p_{ik}^{dis} \leq |p^{dis, max}| \cdot \underline{\delta_{ik}^{dis}}$, $i=0, 1, \dots, N-1$ (DISH. INDICATOR VARIABLE)
- $\delta_{ik}^{ch} + \delta_{ik}^{dis} \leq 1$, $i=0, 1, \dots, N-1$ (MUTUALLY EXCLUSIVE CH. AND DISCH.)

- $x_{i+1k} = x_{ik} + T \eta^{ch} p_{ik}^{ch} - \frac{1}{\eta^{dis}} T p_{ik}^{dis}$, $i=0, 1, \dots, N-1$ (DYNAMICS WITH LOSSES)
- $x_{0k} = x_k$ (STATE MEASUREMENT)

- $x^{\min} \leq x_{ik} \leq x^{\max}, i=0, 1, \dots, N$

- $p^{\text{dis}, \max} \leq p_{ik}^{\text{ess}} \leq p^{\text{ch}, \max}, i=0, 1, \dots, N-1$

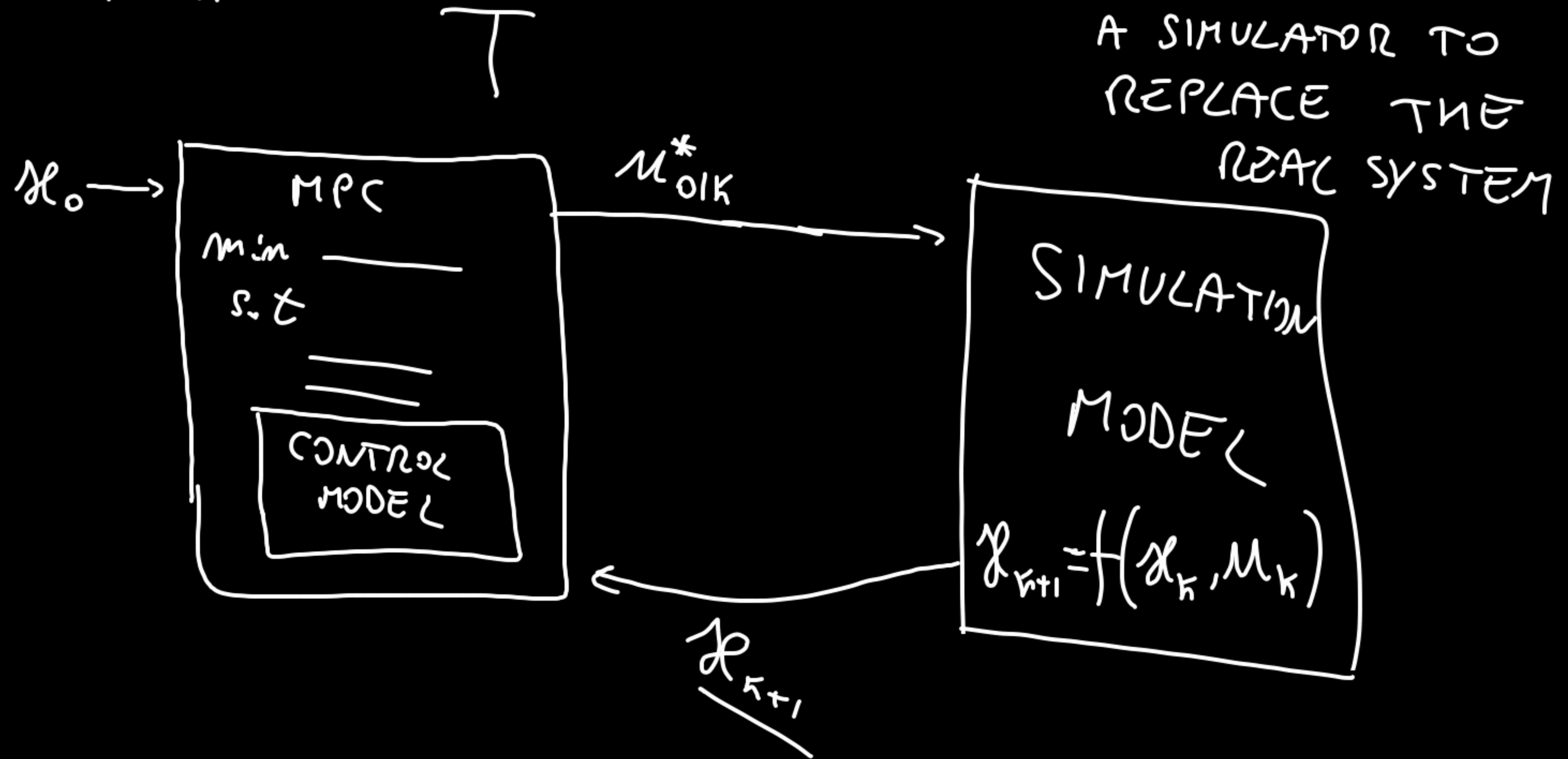
- $p^{\text{POC}, \min} \leq p_{ik}^{\text{POC}} \leq p^{\text{POC}, \max}, i=0, 1, \dots, N-1$

BOX
CONSTRAINTS

- $\left(\begin{array}{l} 0 \leq p_{ik}^{\text{ch}} \leq p^{\text{ch}, \max} \\ 0 \leq p_{ik}^{\text{dis}} \leq p^{\text{dis}, \max} \end{array} \right), i=0, 1, \dots, N-1$

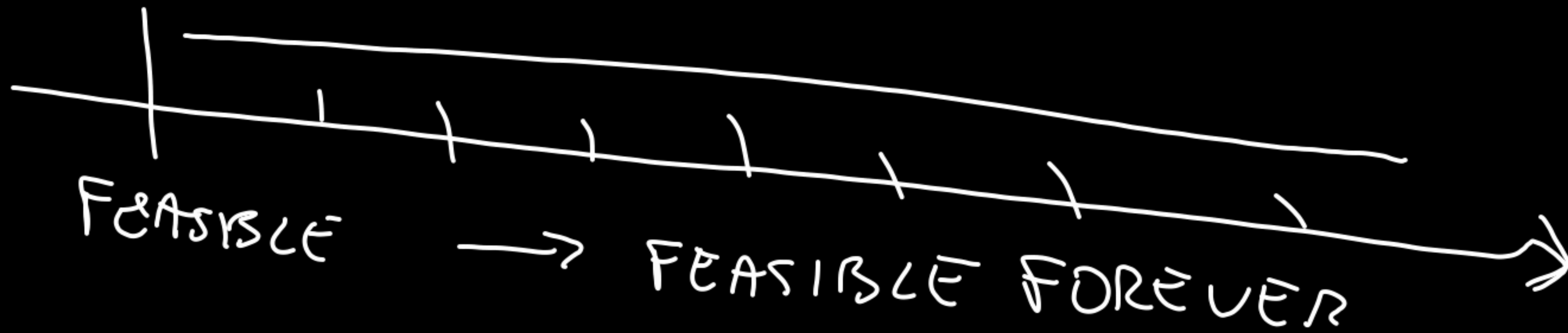
CONTROL MODEL AND SIMULATION MODEL

AT EVERY k



RECURSIVE FEASIBILITY:

DEF. AN MPC ALGORITHM IS RECURSIVE FEASIBLE $\iff$ IF THE MPC ITERATION IS FEASIBLE AT A GIVEN TIME k_0 , THEN IT WILL BE FEASIBLE $\forall k > k_0$.



REFERENCE MPC FORMULATION

$$P_N(x_k) : \min_{u_k} \left\{ \sum_{i=0}^{N-1} \ell(u_{i|k}, x_{i|k}) + V_f(x_{N|k}) \right\}$$

DEF:

$$u_k = \{u_{0|k}, \dots, u_{N-1|k}\}$$

s.t.

$$\left[\begin{array}{l} x_{i+1|k} = f(x_{i|k}, u_{i|k}) ; i = 0, 1, \dots, N-1 \\ x_{0|k} = \underline{x_k} \text{ (INITIAL STATE)} \end{array} \right] \text{MODEL}$$

$$\left[\begin{array}{l} u_{i|k} \in U ; i = 0, 1, 2, \dots, N-1 \\ x_{i|k} \in X, i = 0, 1, \dots, N \end{array} \right] \text{ADMISSIBLE CONTROLS/STATES}$$

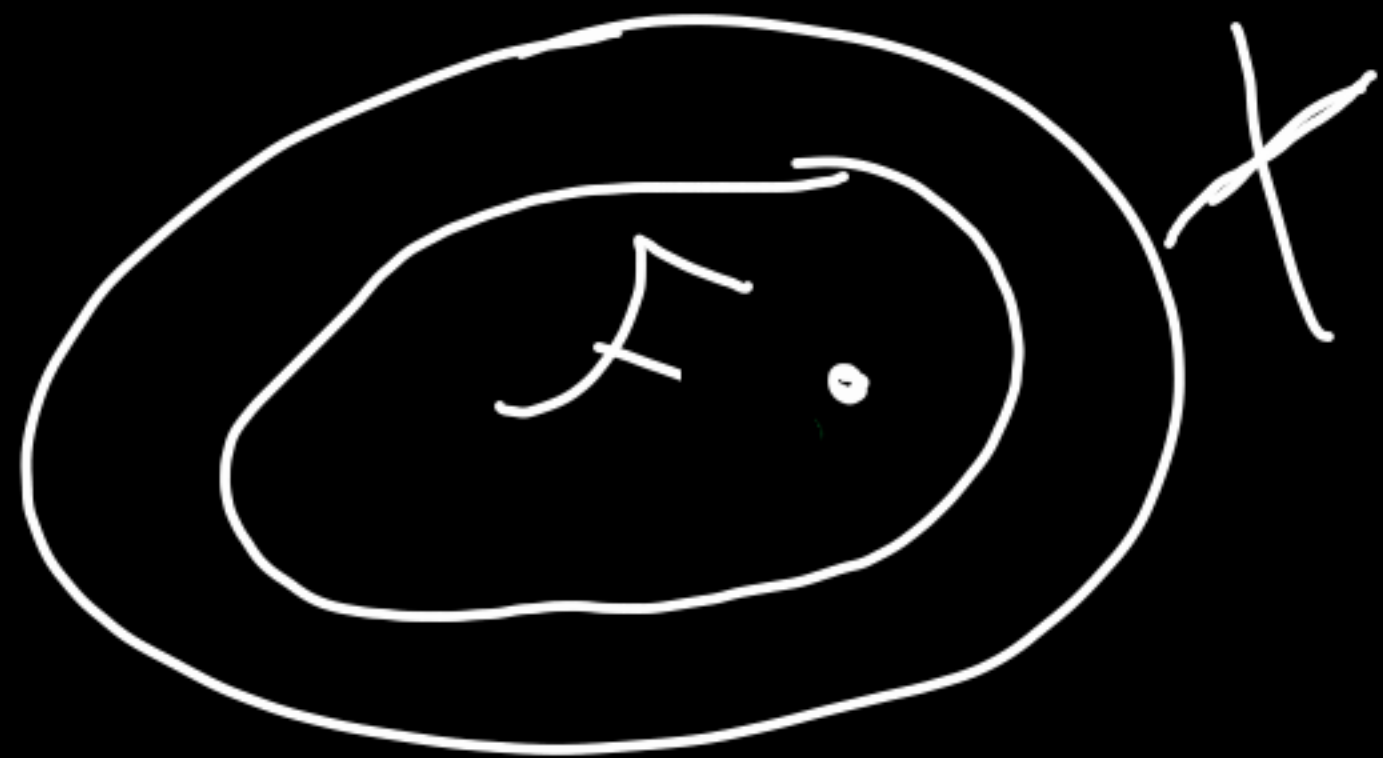
$$\left[\boxed{x_{N|k} \in X_f} \rightarrow \text{TERMINAL SET} \right] \text{TERMINAL CONSTRAINT}$$



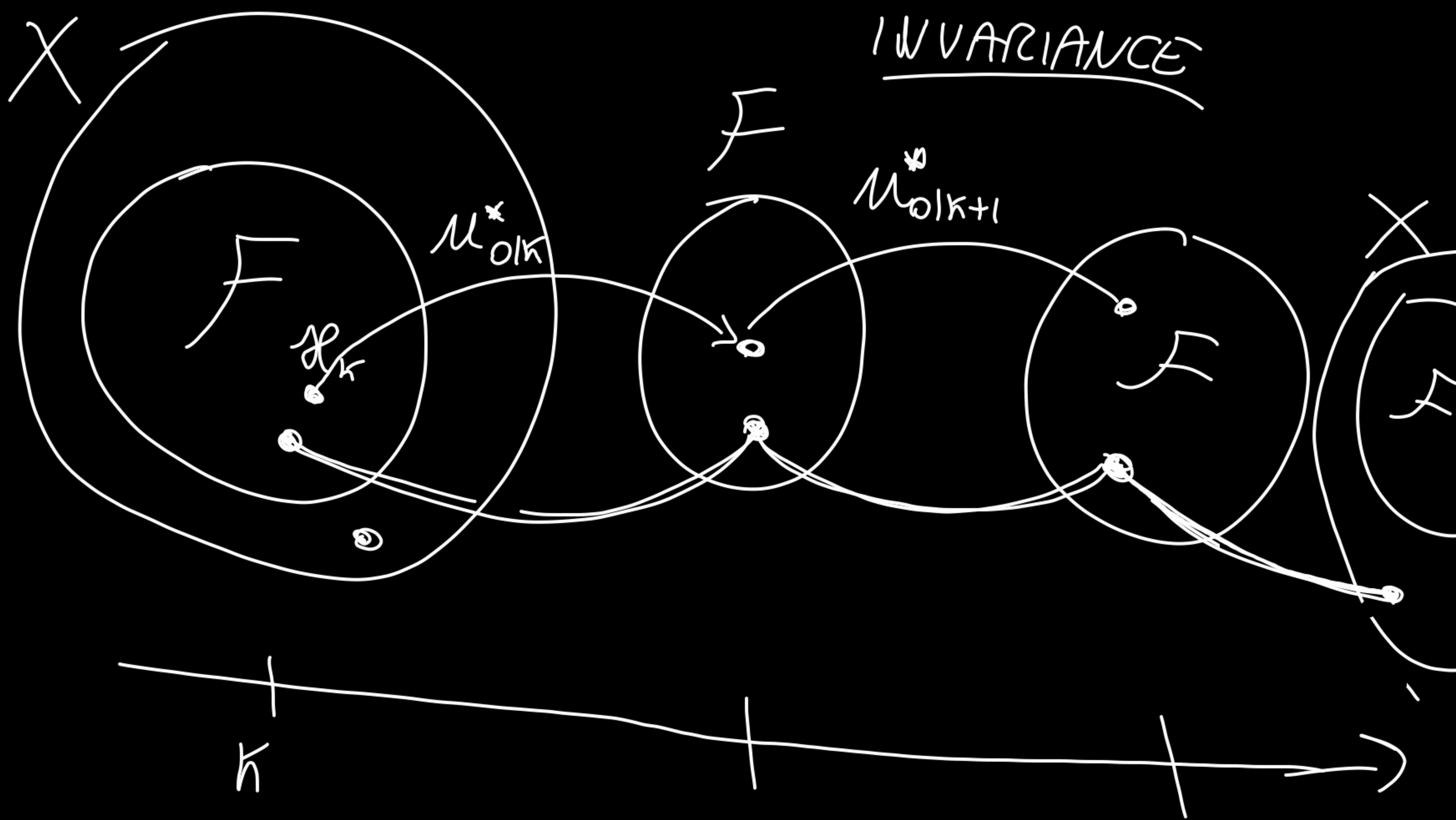
PROF REC. FEAS. DEFINITION $\rightarrow$ FACTS

FEASIBLE SET: SET OF STATES FOR WHICH THE
MPC ITERATION IS FEASIBLE.

$$\mathcal{F} = \{x \in X : P_N(x) \text{ IS FEASIBLE}\}$$



INVARIANCE



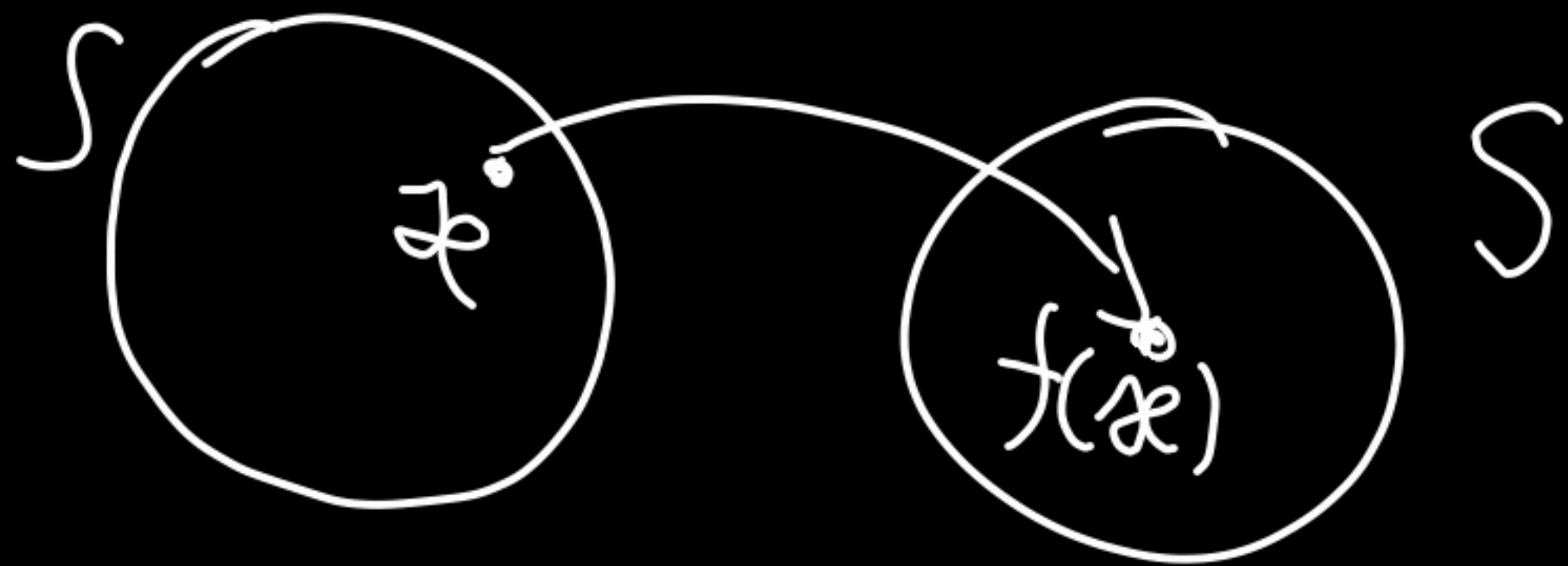
INVARIANCE AND CONTROLLED INVARIANCE

• CONSIDER A SYSTEM (AUTONOMOUS) $x_{k+1} = f(x_k)$

DEF. (INVARIANCE)

A SET S IS CALLED INVARIANT WITH RESPECT TO THE SYSTEM $x_{k+1} = f(x_k)$ IF AND ONLY IF:

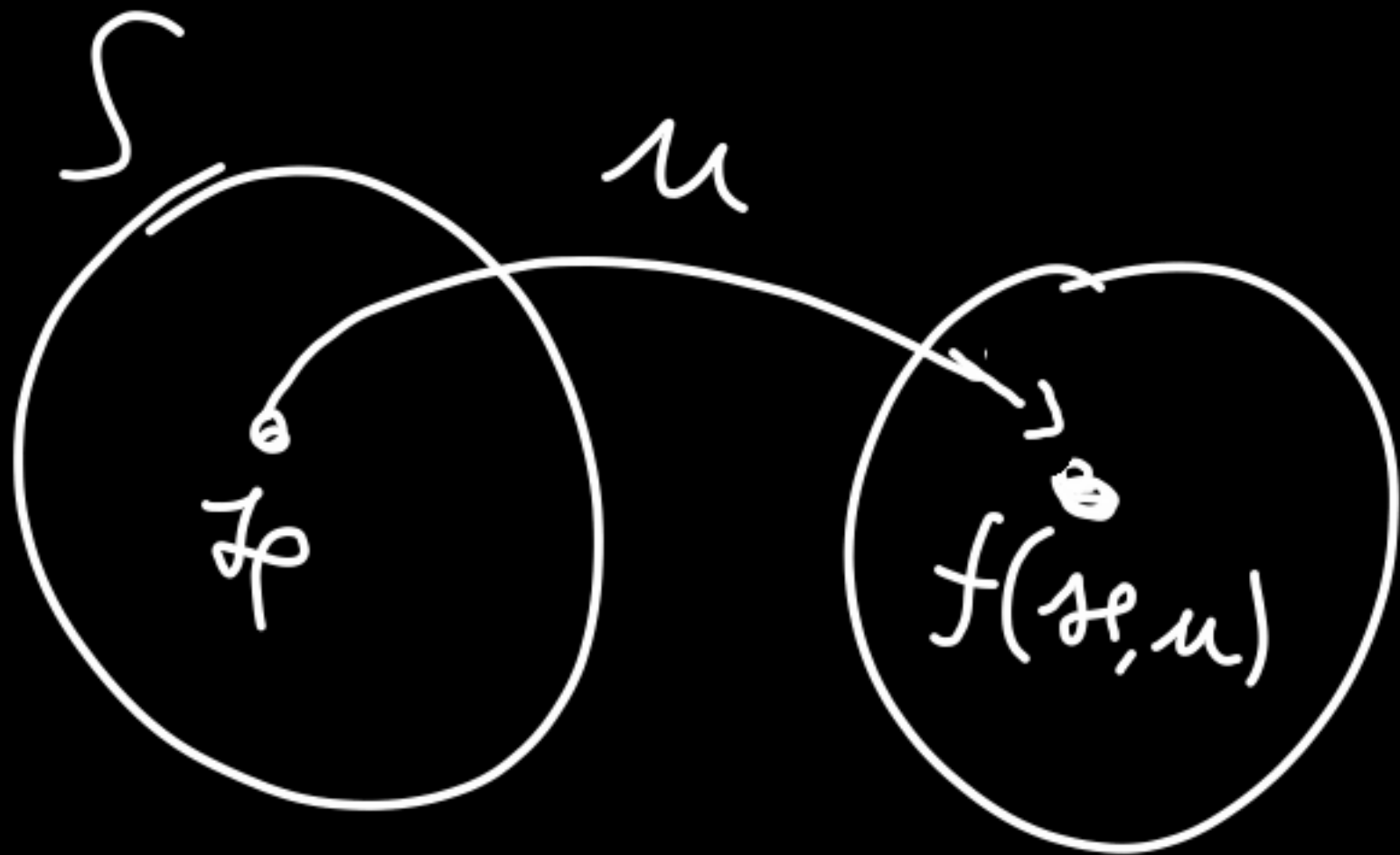
$$\forall x \in S \longrightarrow f(x) \in S$$



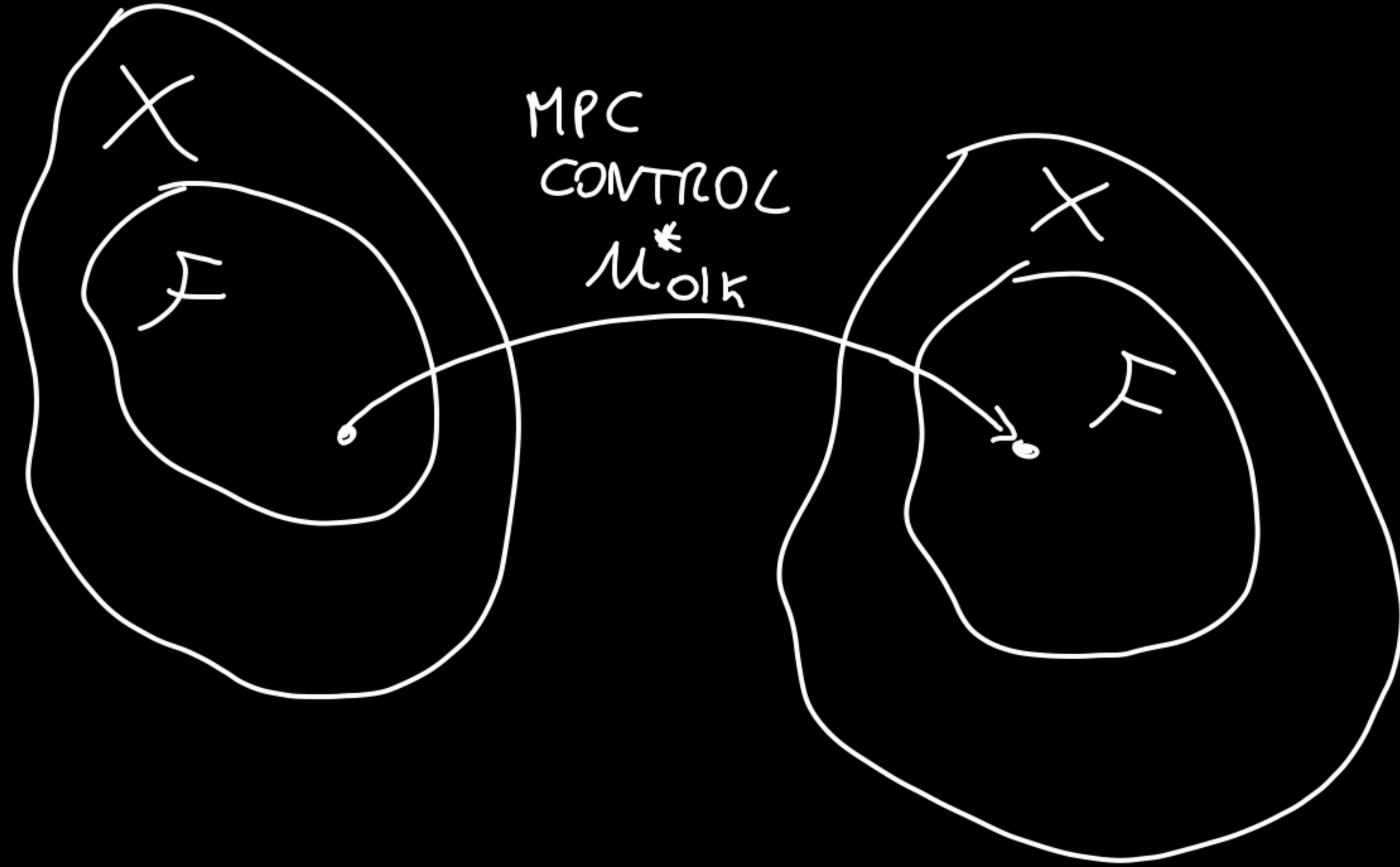
DEF. CONTROLLED INVARIANCE :

A SET IS CONTROLLED INVARIANT W.R.T $x_{k+1} = f(x_k, u_k)$
IF AND ONLY IF :

$$\forall x \in S, \exists u : f(x, u) \in S$$



RECURSIVE FEASIBLE MPC AND CONTROLLED INVARIANCE

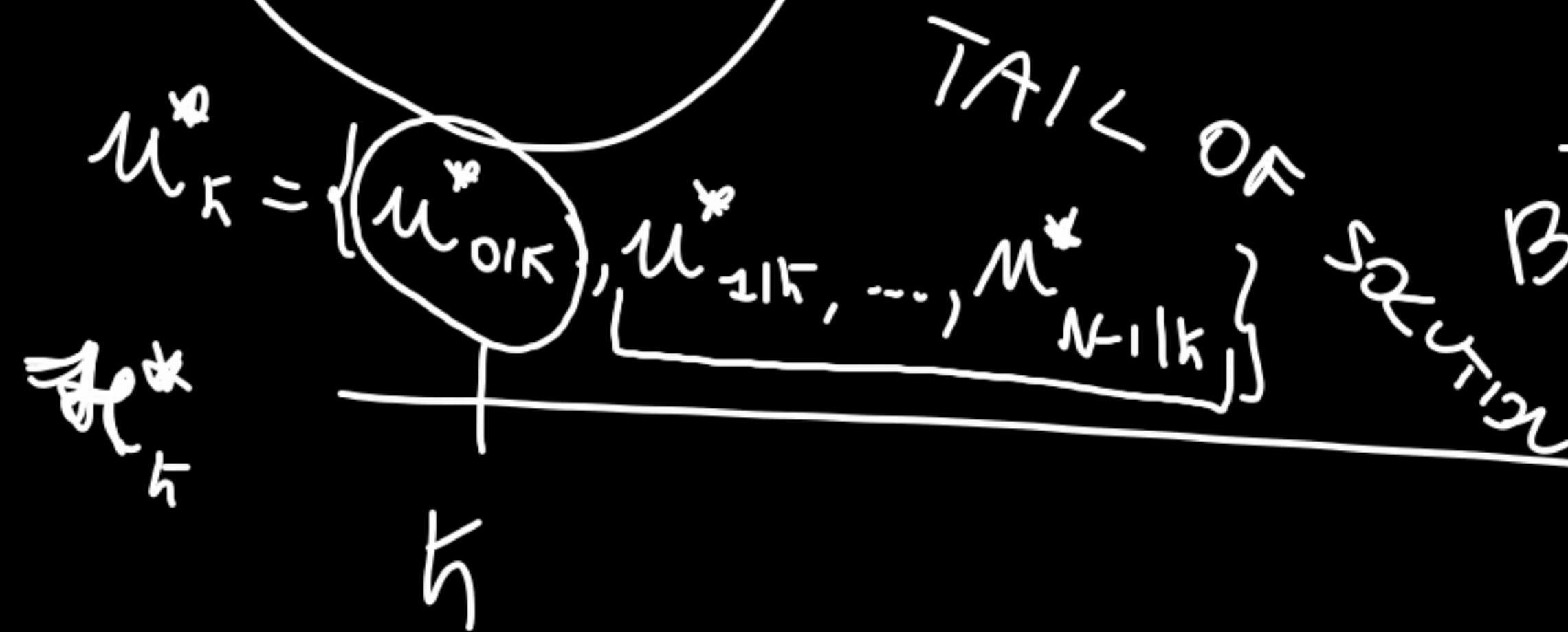


FOR A REC. FEAS.
MPC ALGORIT.,
THE MPC CONTROL
MAKES F CONTROL-
LED INVARIANT.

GIVEN AN MPC PROBLEM, WHAT TO DO TO PROVE IT IS REC. FEASIBLE?



WE MUST PROVE THAT
THERE IS AT LEAST ONE
FEASIBLE SOLUTION AT $k+1$.



CONSTRUCTIVE APPROACH: WE
BUILD A TENTATIVE SOLUTION AND
WE SHOW IT SATISFIES ALL CONSTR.

